

Vootakoti Devaharshini

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EDUCATION

Vignan's Institute of Information Technology

B.Tech in Artificial Intelligence Engineering — CGPA: 9.1

Indian Institute of Technology Madras

BS in Data Science and Applications

Visakhapatnam, India

2023 – 2027

Online Degree Program

2023 – 2027

PROJECTS

Retrieval-Augmented AI Assistant | *FastAPI, LangChain, LangGraph, ChromaDB, Next.js* | [GitRepo](#)

- Engineered a scalable RAG-based AI assistant using LangChain, LangGraph, ChromaDB, and FastAPI for context-aware document question answering
- Implemented PDF ingestion, semantic chunking, vector embeddings, and similarity-based retrieval pipelines to reduce hallucinations and improve response accuracy
- Built full-stack architecture with conversational memory, prompt optimization, Ollama-based local LLM inference, and real-time Next.js frontend interaction

Autonomous AI QA Testing Agent | *LangGraph, Playwright, FastAPI, Ollama* | [GitRepo](#)

- Built an autonomous AI-powered browser testing agent capable of planning, executing, and recovering browser workflows using LangGraph and Playwright
- Implemented self-healing selector recovery, fault injection testing, reflection-based execution analysis, and memory-driven recovery systems for resilient QA automation
- Integrated Ollama-based local LLM reasoning with DOM-aware observation pipelines for adaptive browser interaction and workflow generation

Music Genre Classification using Deep Learning | *PyTorch, CNN, Librosa* | [GitRepo](#) | [Notebook](#)

- Developed an audio classification pipeline using Mel-spectrogram and MFCC feature extraction techniques for music genre prediction
- Implemented CNN-based deep learning models with spectrogram inputs to capture temporal and frequency-based audio patterns
- Applied data augmentation, regularization, and hyperparameter tuning techniques to improve robustness on noisy audio datasets

Placement Portal Application | *Flask, Vue.js, PostgreSQL, Redis* | [GitRepo](#) | [Demo](#)

- Developed a full-stack placement management platform supporting role-based access control for students and administrators
- Designed RESTful APIs with JWT-based authentication workflows using Flask, SQLAlchemy ORM, and PostgreSQL
- Integrated Redis caching and dynamic Vue.js dashboards for optimized session management and application tracking workflows

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL, C

AI & Machine Learning: Scikit-learn, PyTorch, TensorFlow, NLP, Deep Learning, RAG Pipelines, Prompt Engineering, LangChain, LangGraph

Frameworks & Backend: FastAPI, Flask, REST APIs, SQLAlchemy, JWT Authentication, Role-Based Access Control

Frontend & Web Technologies: Next.js, Vue.js, HTML, CSS, Bootstrap, Jinja2

Databases & Infrastructure: PostgreSQL, SQLite, Redis, ChromaDB, Vector Databases, Docker

Developer Tools: Git, GitHub, Playwright, Ollama, Google Colab, VS Code

Coursework: Data Structures & Algorithms, DBMS, Operating Systems, OOPs, Computer Networks

Certifications: Python (Infosys Springboard), JavaScript Essentials (Cisco), HTML & CSS (Cisco)

EXPERIENCE

Deloitte Data Analytics Virtual Experience Program

Forage

Remote

2025

- Completed a virtual data analytics simulation involving business dataset analysis and insight generation
- Performed data interpretation and created visual summaries to support business decision-making scenarios